

STT 465

Bayesian Data Analysis

Lecture 04: Single-parameter Models and Posterior Summaries I

Guanyu Hu

Department of Statistics and Probability
Michigan State University

- A working understanding of probability theory
 - Probability $(\Omega, \mathcal{F}, P)$
 - Conditional probability
 - Independence
 - Bayes' theorem
 - Random variable X
 - Probability functions: c.d.f, p.m.f., and p.d.f
 - Expectation, variance, and covariance
 - Discrete and continuous common distributions
 - Joint, conditional, and marginal probability functions

Learning Goals

- Study typical one/single-parameter Bayesian models
 - Bernoulli or binomial model
 - Bayesian inference
 - Posterior summaries

The Model

- Sample $\mathbf{x}$
 - Observed data: $\mathbf{x} = (x_1, \dots, x_n), x_i \in \{0, 1\}$
 - A collection of i.i.d. observations from a population
 - Sample size n and sample sum $x_{\cdot} = \sum_{i=1}^n x_i$
- Population p_X
 - Bernoulli r.v. $X_i | \omega \sim \text{Bern}(\omega)$ or
Binomial r.v. $X_{\cdot} | n, \omega \sim \text{Bin}(n, \omega)$
 - p.m.f.: $p(X_i = x_i) = \omega^{x_i} (1 - \omega)^{1-x_i}$
 - Moments: $E[X_i] = \omega$ and $\text{Var}(X_i) = \omega(1 - \omega)$
- Parameter ω
 - Support: $\omega \in [0, 1]$
 - Interpretation: Probability of “success” in a Bernoulli trial

Example 1

- A treatment is tested on 21 patients and it is found to have a positive effect in 18 patients. What is the probability that the treatment is effective?

$$x_1, \dots, x_n | \omega \sim \text{Bern}(\omega)$$

- Denote X_i as the binary outcome of patient i after the treatment with $X_i = 1$ for effective
- Sample size $n = 21$
- Sample sum $x_{\cdot} = \sum_{i=1}^n x_i = 18$
- Effective rate ω

Example 2

- A study in Germany found that of a 980 placenta previa births, 437 were female. What is the proportion of female births in the population of women with placenta previa?

$$x_1, \dots, x_n | \omega \sim \text{Bern}(\omega)$$

- Denote X_i as the gender of birth i with $X_i = 1$ for female
- Sample size $n = 980$
- Sample sum $x_{\cdot} = \sum_{i=1}^n x_i = 437$
- The female birth proportion ω

Frequentist: Point Estimation

- Method of moments (MOM)

$$\hat{\omega} = \bar{x},$$

where $\bar{x}$ denotes the sample mean $\bar{x} = x_{\cdot}/n$

- Maximum likelihood estimator (MLE)

$$\frac{d \log L(\omega|\mathbf{x})}{d\omega} = 0 \quad \Rightarrow \quad \hat{\omega} = \bar{x},$$

where $L(\omega|\mathbf{x}) = \prod_{i=1}^n \omega^{x_i} (1 - \omega)^{1-x_i} = \omega^{n\bar{x}} (1 - \omega)^{n-n\bar{x}}$

- Uniform minimum variance unbiased estimator (UMVUE)

$$\hat{\omega} = \bar{x}$$

- Examples

- Effective rate $\hat{\omega} = 18/21 = 0.857$
- Female proportion $\hat{\omega} = 437/980 = 0.446$

Frequentist: “Uncertainty” Quantification

- $(1 - \alpha) \times 100\%$ confidence interval
 - CLT: $\bar{X} \sim N(\omega, \omega(1 - \omega)/n)$ as $n \rightarrow \infty$
 - $\frac{\bar{X} - \omega}{\sqrt{\omega(1 - \omega)/n}} \sim N(0, 1)$
 - $P\left(-z^* \leq \frac{\bar{X} - \omega}{\sqrt{\omega(1 - \omega)/n}} \leq z^*\right) = 1 - \alpha$
 - Approximated 95% confidence interval, where $z^* = 1.960$

$$\left[\bar{x} - 1.960\sqrt{\frac{\bar{x}(1 - \bar{x})}{n}}, \bar{x} + 1.960\sqrt{\frac{\bar{x}(1 - \bar{x})}{n}} \right]$$

- More accurate confidence interval:
 - `prop.test(x = x., n = n, conf.level = 1 -  $\alpha$ )$conf.int`
 - `binom.test(x = x., n = n, conf.level = 1 -  $\alpha$ )$conf.int`

Frequentist: “Uncertainty” Quantification

- Example 1: Effective rate

- Point estimator $\hat{\omega} = 18/21 = 0.857$
- 95% confidence interval
 - By hand: [0.707, 1.007]
 - `prop.test()`: [0.626, 0.962]
 - `binom.test()`: [0.637, 0.970]

- Example 2: Female proportion

- Point estimator $\hat{\omega} = 437/980 = 0.446$
- 95% confidence interval
 - By hand: [0.415, 0.477]
 - `prop.test()`: [0.415, 0.478]
 - `binom.test()`: [0.414, 0.478]
- If ω lies outside the given 95% interval, then obtaining $\hat{\omega}$ in the interval has a probability of 5% at most by chance

Frequentist: Hypothesis Testing

- Example 1: Is the effective rate greater than 0.5?
 - Identify the null and alternative hypotheses

$$H_0 : \omega \leq 0.5 \text{ vs. } H_1 : \omega > 0.5$$

- Specify a test statistics and compute the p -value under H_0
 - `prop.test(x = x., n = n, alternative = "greater")`
 - `binom.test(x = x., n = n, alternative = "greater")`
 - The resulting p -value = 0.001
- Specify a significance level $\alpha = 0.05$
- Conclude that H_0 should be rejected

Frequentist: Hypothesis Testing

- Example 2: Is the male:female ratio at birth is significantly higher (i.e. ω is significantly lower) in women with placenta previa than in those without placenta previa?
- A control study in Germany found that of a 820 birth without placenta previa, 398 were female
 - Identify the null and alternative hypotheses

$$H_0 : \omega_x \geq \omega_y \text{ vs. } H_1 : \omega_x < \omega_y$$

- Specify a test statistics and compute the p -value under H_0
 - `prop.test(x = c(x., y.), n = c(n_x, n_y), alternative = "less")`
 - The resulting p -value = 0.052
- Specify a significance level $\alpha = 0.05$
- Conclude that H_0 cannot be rejected

- Assume data sampling is i.i.d.
- Full data likelihood

$$\begin{aligned}f(\mathbf{x}|\omega) &= f(x_1, \dots, x_n|\omega) \\&= \prod_{i=1}^n f(x_i|\omega) \\&= \prod_{i=1}^n \omega^{x_i} (1 - \omega)^{1-x_i} \\&= \omega^{\sum_{i=1}^n x_i} (1 - \omega)^{n - \sum_{i=1}^n x_i} \\&= \omega^{x_{\cdot}} (1 - \omega)^{n - x_{\cdot}}.\end{aligned}$$

Prior

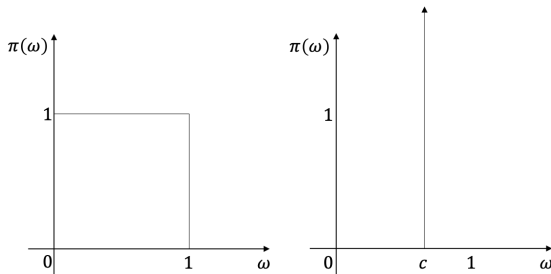
- Two extreme cases

- Uniform prior $\omega \sim \text{Unif}(0, 1)$

$$\pi(\omega) = 1, 0 \leq \omega \leq 1$$

- Point mass prior $\omega = c, c \in [0, 1]$

$$\pi(\omega) = \mathbb{I}(\omega = c)$$



Posterior Inference

- Data likelihood: $f(\mathbf{x}|\omega) = \omega^{x_{\cdot}}(1 - \omega)^{n - x_{\cdot}}$.
- Prior distribution: $\pi(\omega) = 1, 0 \leq \omega \leq 1$
- Posterior
 - Bayes' theorem

$$p(\theta|\mathbf{x}) = \frac{p(\theta, \mathbf{x})}{p(\mathbf{x})} = \frac{p(\mathbf{x}|\theta)p(\theta)}{\int p(\mathbf{x}|\theta)p(\theta)d\theta} \propto p(\mathbf{x}|\theta)p(\theta)$$

- Posterior distribution

$$\pi(\omega|\mathbf{x}) \propto f(\mathbf{x}|\omega)\pi(\omega) = \omega^{x_{\cdot}}(1 - \omega)^{n - x_{\cdot}}.$$

- Beta component

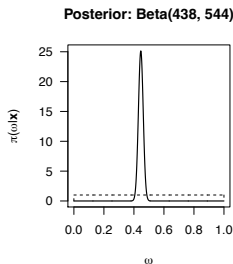
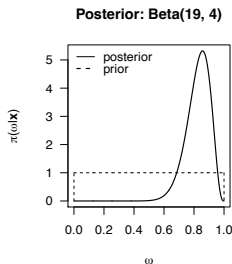
If $p(\theta) \propto \theta^a(1 - \theta)^b$, then $\theta \sim \text{Beta}(a + 1, b + 1)$

Posterior

- Posterior: $\omega|\mathbf{x} \sim \text{Beta}(1 + x_., 1 + n - x_.)$

$$\pi(\omega|\mathbf{x}) = \frac{\Gamma(2 + n)}{\Gamma(1 + x_.)\Gamma(1 + n - x_.)} \omega^{x_.} (1 - \omega)^{n - x_..}$$

- Example 1: $\omega|\mathbf{x} \sim \text{Beta}(1 + 18, 1 + 21 - 18)$
- Example 2: $\omega|\mathbf{x} \sim \text{Beta}(1 + 437, 1 + 980 - 437)$
- Make any inference from posterior distributions



Bernoulli-beta Model

- Beta prior: $\omega \sim \text{Beta}(a, b)$

$$\pi(\omega) = \frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)} \omega^{a-1} (1-\omega)^{b-1}$$

- Reduce to $\omega \sim \text{Unif}(0, 1)$ if $a = b = 1$
- Posterior

$$\begin{aligned}\pi(\omega|\mathbf{x}) &\propto f(\mathbf{x}|\omega)\pi(\omega) \\ &= \omega^{x_{\cdot}} (1-\omega)^{n-x_{\cdot}} \frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)} \omega^{a-1} (1-\omega)^{b-1} \\ &= \frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)} \omega^{a+x_{\cdot}-1} (1-\omega)^{b+n-x_{\cdot}-1} \\ &\propto \omega^{a+x_{\cdot}-1} (1-\omega)^{b+n-x_{\cdot}-1}\end{aligned}$$

Bernoulli-beta Model

- Posterior: $\omega|\mathbf{x} \sim \text{Beta}(a + x_., b + n - x_.)$

$$\pi(\omega|\mathbf{x}) = \frac{\Gamma(a + b + n)}{\Gamma(a + x_.)\Gamma(b + n - x_.)} \omega^{a+x_.-1} (1 - \omega)^{b+n-x_.-1}$$

- Bernoulli-beta model

$$x_1, \dots, x_n | \omega \sim \text{Bern}(\omega)$$

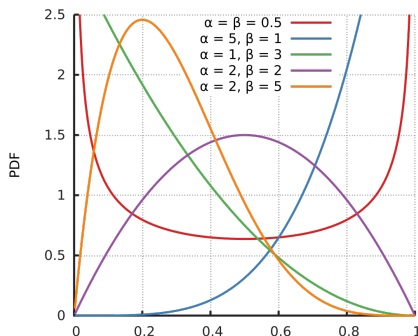
$$\omega \sim \text{Beta}(a, b)$$

$$\omega | x_1, \dots, x_n \sim \text{Beta}(a + x_., b + n - x_.)$$

- Conjugate prior: The prior $\pi(\theta)$ and posterior $\pi(\theta|\mathbf{x})$ are called conjugate distributions, if they are in the same family of distributions

Prior Choices

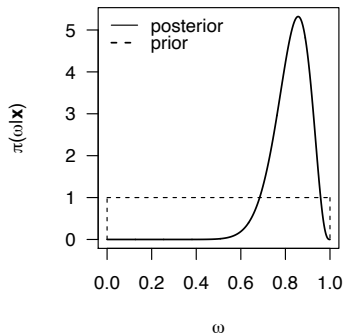
- Beta prior: $\omega \sim \text{Beta}(a, b)$ with $E[\omega] = a/(a + b)$
 - Symmetric if $a = b$
 - Unimodal if $a, b > 1$
 - A spike at one end (J or S-shaped) if $a \leq 1$ or $b \leq 1$
 - Two spikes at each end (U-shaped) if $a, b < 1$



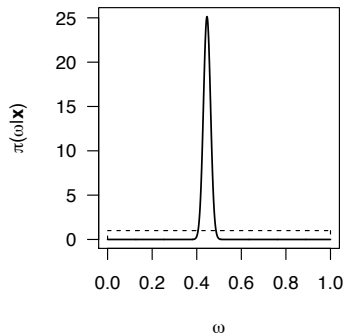
Posterior Distributions

- Make inferences from the posterior distribution

Posterior: Beta(19, 4)



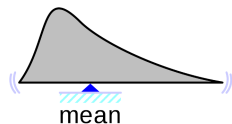
Posterior: Beta(438, 544)



Distribution Descriptors

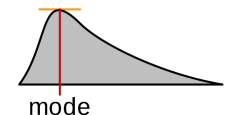
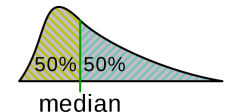
- Center

- Mean: The balanced value
- Median: The middle value
- Mode: The most likely value



- Spread

- Variance: Value-based
- Inter-quartile range (IQR): Position-based
- Median absolute deviation (MAD): Hybrid



Distribution Descriptors

- Continuous common distributions

	Mean	Median	Mode	Variance
$\text{Unif}(a, b)$	$\frac{a+b}{2}$	$\frac{a+b}{2}$	Any value	$\frac{(b-a)^2}{12}$
$\text{Beta}(a, b)$	$\frac{a}{a+b}$	$\frac{a - \frac{1}{3}}{a+b - \frac{2}{3}}$	$\frac{a-1}{a+b-2}$	$\frac{ab}{(a+b)^2(a+b+1)}$
$\text{Exp}(\lambda)$	$\frac{1}{\lambda}$	$\frac{\log 2}{\lambda}$	0	$\frac{1}{\lambda^2}$
$\text{Ga}(a, b)$	$\frac{a}{b}$	No closed form	$\frac{a-1}{b}$	$\frac{a}{b^2}$
$\text{IG}(a, b)$	$\frac{b}{a-1}$	No closed form	$\frac{b}{a+1}$	$\frac{b^2}{(a-1)^2(a-2)}$
$N(\mu, \sigma^2)$	μ	μ	μ	σ^2
$\text{LN}(\mu, \sigma^2)$	$\exp\left(\mu + \frac{\sigma^2}{2}\right)$	$\exp(\mu)$	$\exp(\mu - \sigma^2)$	$(\exp(\sigma^2) - 1) \exp(2\mu + \sigma^2)$
$t(\nu, \mu, \sigma)$	μ	μ	μ	$\sigma^2 \frac{\nu}{\nu-2}$

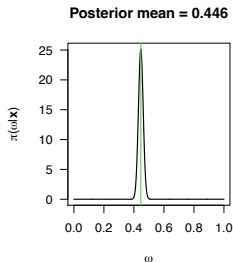
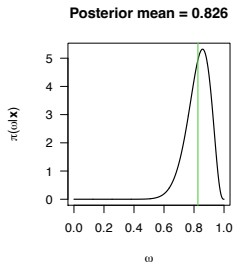
Bayesian: Point Estimation

- Posterior mean

$$\hat{\theta} = E[\theta|\mathbf{x}]$$

- Bernoulli-beta model and examples

$$\hat{\omega} = \frac{a + x.}{a + b + n}$$



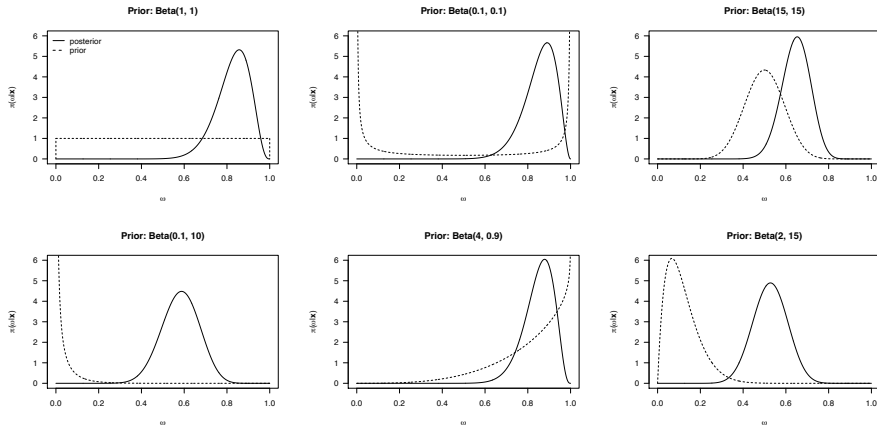
- Posterior mean is a weighted average of prior mean and sample mean

$$\begin{aligned} E[\omega|\mathbf{x}] &= \frac{a + x.}{a + b + n} \\ &= \frac{a + b}{a + b + n} \frac{a}{a + b} + \frac{n}{a + b + n} \frac{x.}{n} \\ &= \frac{a + b}{a + b + n} E[\omega] + \frac{n}{a + b + n} \bar{x} \end{aligned}$$

- Posterior is a compromise between prior and data depending on the relative weight of $a + b$ w.r.t. n
- With a and b fixed, $n \rightarrow \infty \Rightarrow E[\omega|\mathbf{x}] \rightarrow \bar{x}$
- With n fixed, $a + b \rightarrow \infty \Rightarrow E[\omega|\mathbf{x}] \rightarrow E[\omega]$

Sensitivity Analysis

- Posterior distribution under different choices of a and b



Bayesian: Point Estimation

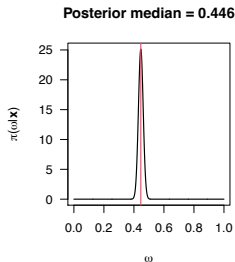
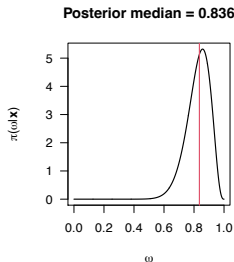
- Posterior median

$$\hat{\theta} = F_{\theta|x}^{-1}(0.5)$$

- Bernoulli-beta model and examples

$$\hat{\omega} \approx \frac{a + x. - 1/3}{a + b + n - 2/3} \text{ if } 0 < x. < n$$

`qbeta(p = 0.5, shape1 = a + x., shape2 = b + n - x.)`



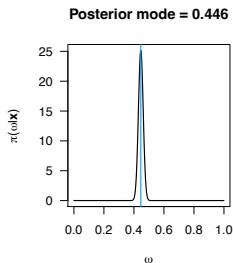
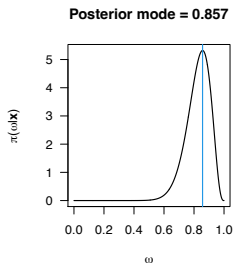
Bayesian: Point Estimation

- Posterior mode

$$\hat{\theta} = \arg \max_{\theta} \pi(\theta | \mathbf{x})$$

- Bernoulli-beta model and examples

$$\hat{\omega} = \frac{a + x_{\cdot} - 1}{a + b + n - 2} \text{ if } 0 < x_{\cdot} < n$$



- Also known as *maximum a posteriori* (MAP) estimate
- Posterior mode is MLE if $\pi(\theta) \propto 1$

$$\begin{aligned}\hat{\theta} &= \arg \max_{\theta} \pi(\theta | \mathbf{x}) \\ &= \arg \max_{\theta} f(\mathbf{x} | \theta) \pi(\theta) \\ &= \arg \max_{\theta} f(\mathbf{x} | \theta) \\ &= \arg \max_{\theta} L(\theta | \mathbf{x})\end{aligned}$$

- Frequentist is a special case of Bayesian when $\pi(\theta) \propto 1$

Bayesian: Uncertainty Quantification

- Posterior variance

$$\text{Var}(\theta|\mathbf{x})$$

- Bernoulli-beta model

$$\begin{aligned}\text{Var}(\omega|\mathbf{x}) &= \frac{(a + x_{\cdot})(b + n - x_{\cdot})}{(a + b + n)^2(a + b + n + 1)} \\ &= \frac{a + x_{\cdot}}{a + b + n} \frac{b + n - x_{\cdot}}{a + b + n} \frac{1}{a + b + n + 1} \\ &= \frac{a + x_{\cdot}}{a + b + n} \left(1 - \frac{a + x_{\cdot}}{a + b + n}\right) \frac{1}{a + b + n + 1} \\ &= \frac{\text{E}[\omega|\mathbf{x}](1 - \text{E}[\omega|\mathbf{x}])}{a + b + n + 1}\end{aligned}$$

- With a and b fixed, $n \rightarrow \infty \Rightarrow \text{Var}(\omega|\mathbf{x}) \rightarrow 0$
- $\text{Var}(\omega|\mathbf{x}) \leq \text{Var}(\omega)$

Discussion

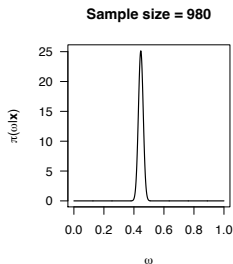
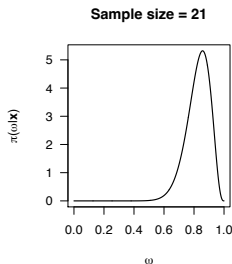
- A posterior distribution is approximately normal for large n

$$\theta|\mathbf{x} \sim N(E[\theta|\mathbf{x}], \text{Var}(\theta|\mathbf{x})), \text{ as } n \rightarrow \infty$$

- Approximation is more accurate on a transformed scale

- Logit transformation $\log \frac{\theta}{1-\theta}$ for $\theta \in [0, 1]$
- Log transformation $\log \theta$ for $\theta \in \mathbb{R}_+$

- Frequentist is a special case of Bayesian when $n \rightarrow \infty$



Bayesian: Uncertainty Quantification

- $(1 - \alpha) \times 100\%$ credible interval

$$P(\hat{\theta}_{\text{lwr}} \leq \theta \leq \hat{\theta}_{\text{upp}} | \mathbf{x}) = 1 - \alpha$$

- Equal-tailed interval (central interval)

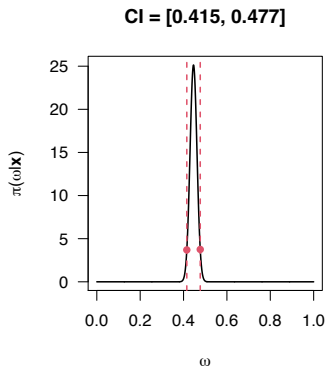
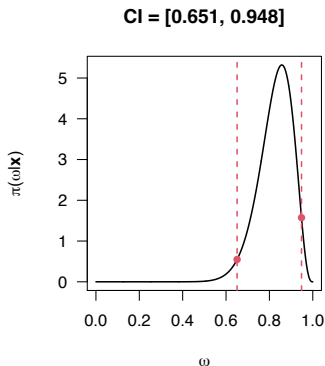
$$\hat{\theta}_{\text{lwr}} = F_{\theta | \mathbf{x}}^{-1}(\alpha/2) \text{ and } \hat{\theta}_{\text{upp}} = F_{\theta | \mathbf{x}}^{-1}(1 - \alpha/2)$$

- Credible interval (CI) by default
- Bernoulli-beta model

```
qbeta(p =  $\alpha/2$ , shape1 =  $a + x.$ , shape2 =  $b + n - x.$ )  
qbeta(p =  $1 - \alpha/2$ , shape1 =  $a + x.$ , shape2 =  $b + n - x.$ )
```

Bayesian: Uncertainty Quantification

- $(1 - \alpha) \times 100\%$ credible interval
 - Examples (95% CI)



- Some points outside of the CI have a higher posterior density than some points within the CI

Bayesian: Uncertainty Quantification

- $(1 - \alpha) \times 100\%$ credible interval

$$P(\hat{\theta}_{\text{lwr}} \leq \theta \leq \hat{\theta}_{\text{upp}} | \mathbf{x}) = 1 - \alpha$$

- Highest posterior density interval (HPDI): a set H on the parameter space Θ that satisfies

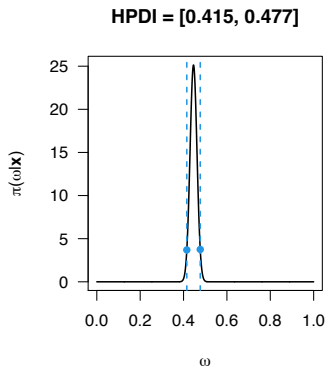
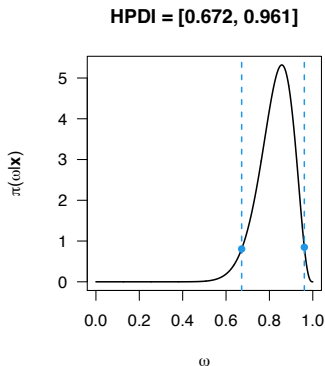
$$f(\theta | \mathbf{x}) \geq f(\theta' | \mathbf{x}), \forall \theta \in H, \theta' \in \Theta \setminus H$$

- Shortest length among all $(1 - \alpha) \times 100\%$ credible intervals
- Bernoulli-beta model

```
require("HDInterval")  
set.seed(1)  
hdi(rbeta(10000, shape1 = a + x., shape2 = b + n - x.),  
    credMass = 1 -  $\alpha$ )
```


Bayesian: Uncertainty Quantification

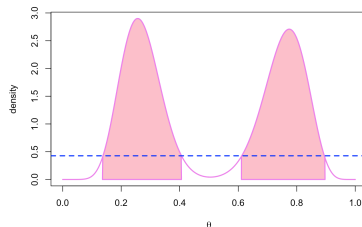
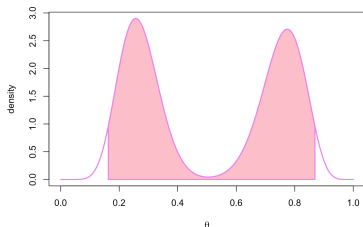
- $(1 - \alpha) \times 100\%$ credible interval
 - Examples (95% HPDI)



- All points within the HPDI have a higher posterior density than those outside of the HPDI

Discussion

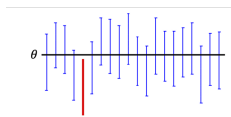
- CI = HPDI for symmetric, unimodal, and concave distributions
- HPDI is always no wider than CI
- CI is an interval, while HPDI is not necessarily an interval



- Credible region and highest posterior density region for multiple-parameter space

Discussion

- Credible interval $\neq$ confidence interval
- Credible interval
 - Parameter θ is random and x is fixed
 - $\hat{\theta}_{lwr}$ and $\hat{\theta}_{upp}$ are known constants
 - θ is inside the credible interval with probability $(1 - \alpha) \times 100\%$
- Confidence interval
 - Parameter θ is fixed
 - $\hat{\theta}_{lwr}$ and $\hat{\theta}_{upp}$ are random
 - With a large number of repeated samples, $(1 - \alpha) \times 100\%$ probability of such calculated confidence intervals would include θ



- “Safety and efficacy of the BNT162b2 mRNA COVID-19 vaccine,” *The New England Journal of Medicine*, 2021 by Polack *et al.*
- One of the main statistical evidences to support that the Pfizer vaccine was 95% effective in preventing COVID-19
- The 95% CI of the vaccine efficacy (VE) is [90.3, 97.6]

Analysis of the first primary efficacy end point included participants who received the vaccine or placebo as randomly assigned, had no evidence of infection within 7 days after the second dose, and had no major protocol deviations (the population that could be evaluated). Vaccine efficacy was estimated by $100 \times (1 - \text{IRR})$, where IRR is the calculated ratio of confirmed cases of Covid-19 illness per 1000 person-years of follow-up in the active vaccine group to the corresponding illness rate in the placebo group. The 95.0% credible interval for vaccine efficacy and the probability of vaccine efficacy greater than 30% were calculated with the use of a Bayesian beta-binomial model. The final analysis uses a success boundary of 98.6% for probability of vaccine efficacy greater than 30% to compensate for the interim analysis and to control the overall type 1 error rate at 2.5%. Moreover, primary and secondary efficacy end points are evaluated sequentially to control the familywise type 1 error rate at 2.5%. Descriptive analyses (estimates of vaccine efficacy and 95% confidence intervals) are provided for key subgroups.

- Credible interval $\neq$ confidence interval

use antipyretic or pain medication (28% after dose 1; 45% after dose 2) than older vaccine recipients (20% after dose 1; 38% after dose 2), and placebo recipients were less likely (10 to 14%) second dose were observed among vaccine recipients and 162 among placebo recipients. This case split corresponds to 95.0% vaccine efficacy (95% confidence interval [CI], 90.3 to 97.6; Ta-

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CONFIDENCE INTERVAL

is NOT

SAFETY AND EFFICACY OF THE BNT162B2 VACCINE

credible interval

Table 2. Vaccine Efficacy against Covid-19 at Least 7 days after the Second Dose.*

Efficacy End Point	BNT162b2		Placebo		Vaccine Efficacy, % (95% Credible Interval)§	Posterior Probability (Vaccine Efficacy >30%)§
	No. of Cases	Surveillance Time (n)†	No. of Cases	Surveillance Time (n)†		
	(N=18,198)		(N=18,325)			
Covid-19 occurrence at least 7 days after the second dose in participants without evidence of infection	8	2,214 (17,411)	162	2,222 (17,511)	95.0 (90.3–97.6)	>0.9999

Example 1

- A treatment is tested on 21 patients and it is found to have a positive effect in 18 patients. What is the probability that the effective rate is greater than 0.5?

Example 1

- Model formulation: $X_i \sim \text{Bern}(\omega)$
- Bayesian inference
 - Data likelihood: $f(\mathbf{x}|\omega) = \omega^{x_{\cdot}}(1 - \omega)^{n - x_{\cdot}}$, where $n = 21$ and $x_{\cdot} = 18$
 - Prior: $\omega \sim \text{Beta}(a = 1, b = 1)$
 - Posterior: $\omega|\mathbf{x} \sim \text{Beta}(a + x_{\cdot}, b + n - x_{\cdot})$
- Question: $P(\omega \geq 0.5|\mathbf{x}) = ?$

Example 1

- Exact solutions

```
# Read data information
n <- 21
x_sum <- 18

# Set the prior setting
a <- 1
b <- 1

# Analytical result 0.9996
print(round(1 - pbeta(q = 0.5, shape1 = a + x_sum, shape2 = b + n -
  x_sum), 4))
```


- Frequentist

- Identify the null and alternative hypotheses

$$H_0 : \omega \leq 0.5 \text{ vs. } H_1 : \omega > 0.5$$

- The resulting p -value $P(\mathbf{x}|\omega \leq 0.5) = 0.001$
- Conclude that H_0 (i.e. the effect reate is not greater than 0.5) should be rejected

- Bayesian

- Quantify the uncertainty on the model parameter

$$\omega \sim \text{Beta}(a = 1, b = 1)$$

- The resulting $P(\omega < 0.5|\mathbf{x}) = 1 - 0.9996$
- Very confident that the effective rate is greater than 0.5

Example 2

- A study in Germany found that of a 980 placenta previa births, 437 were female. In the corresponding control study, there were 398 female births out of a population of 820 women without placenta previa. What is the probability that the male:female ratio at birth is higher in women with placenta previa than in those without?

Example 2

- Model formulation: $X_i|\omega_x \sim \text{Bern}(\omega_x)$ and $Y_i|\omega_y \sim \text{Bern}(\omega_y)$
- Bayesian inference
 - Data likelihood:
 $f(\mathbf{x}|\omega_x) = \omega_x^{x_{\cdot}}(1 - \omega_x)^{n_x - x_{\cdot}}$, where $n_x = 980$ and $x_{\cdot} = 437$
 $f(\mathbf{y}|\omega_y) = \omega_y^{y_{\cdot}}(1 - \omega_y)^{n_y - y_{\cdot}}$, where $n_y = 820$ and $y_{\cdot} = 398$
 - Prior:
 $\omega_x, \omega_y \sim \text{Beta}(a = 1, b = 1)$
 - Posterior:
 $\omega_x|\mathbf{x} \sim \text{Beta}(a + x_{\cdot}, b + n_x - x_{\cdot})$
 $\omega_y|\mathbf{y} \sim \text{Beta}(a + y_{\cdot}, b + n_y - y_{\cdot})$
- Question: $P(\omega_x < \omega_y|\mathbf{x}, \mathbf{y}) = ?$

Monte Carlo Approximations

- Calculate exact posterior quantities can be difficult or impossible at time
- Take a sample of U values from $\pi(\theta|\mathbf{x})$

$$\theta^{(1)}, \dots, \theta^{(U)}$$

to approximate the quantity of interest

- By the law of large numbers
 - Empirical distribution of θ approximates $\pi(\theta|\mathbf{x})$ use histogram or kernel density estimator
 - $E[g(\theta)|\mathbf{x}] \approx \frac{1}{U} \sum_{u=1}^U g(\theta^{(u)})$
 - $P(g(\theta) > c|\mathbf{x}) \approx \frac{1}{U} \sum_{u=1}^U I(g(\theta^{(u)}) > c)$
 - Sample moments/quantiles/functions approximate true moments/quantiles/functions
- Easily extends to higher dimensional settings

Example 1

- Solutions (simulation-based)

```
# Set the algorithm setting
U <- 10000
set.seed(1)
# Simulate samples from the posterior distribution
omega_post <- rbeta(U, shape1 = a + x_sum, shape2 = b + n - x_sum)
# Simulation-based result: 0.9994 (varies with the seed number)
print(round(sum(omega_post >= 0.5)/U, 4))
```

Example 1

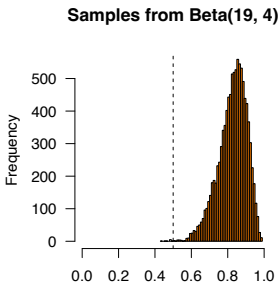
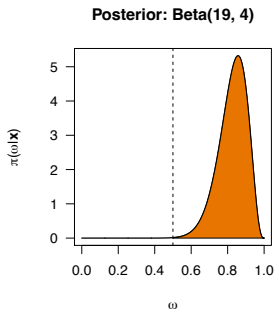
- Solutions

- Analytical

$$P(\omega \geq 0.5 | \mathbf{x}) = 0.9996$$

- Simulation-based

$$P(\omega \geq 0.5 | \mathbf{x}) = 0.9994$$



Example 2

- Solutions (simulation-based)

```
# Read data information
n_x <- 980
x_sum <- 437
n_y <- 820
y_sum <- 398

# Set the prior setting
a <- 1
b <- 1

# Set the algorithm setting
U <- 10000
set.seed(1)
^^I^^I
```

Example 2

- Solutions (simulation-based)

```
# Simulate samples from posterior distributions
omega_x_post <- rbeta(U, shape1 = a + x_sum, shape2 = b + n_x - x_sum)
omega_y_post <- rbeta(U, shape1 = a + y_sum, shape2 = b + n_y - y_sum)

# The answer to Q1 is 0.953
print(round(sum(omega_x_post < omega_y_post)/U, 3))
```


- Frequentist

- Identify the null and alternative hypotheses

$$H_0 : \omega_x \geq \omega_y \text{ vs. } H_1 : \omega_x < \omega_y$$

- The resulting p -value $P(\mathbf{x}, \mathbf{y} | \omega_x \geq \omega_y) = 0.052$
- Conclude that H_0 (i.e. the male:female ratio is higher in women with placenta previa) cannot be rejected

- Bayesian

- Quantify the uncertainty on the model parameter

$$\omega_x, \omega_y \sim \text{Beta}(a = 1, b = 1)$$

- The resulting $P(\omega_x < \omega_y | \mathbf{x}, \mathbf{y}) = 0.953$
- Very confident that the male:female ratio is higher in women with placenta previa

Bayesian: Hypothesis Testing

- Identify the null and alternative hypotheses

$$H_0 : \theta \in A \text{ vs. } H_1 : \theta \in \Theta \setminus A$$

- Formulate the model and compute the posterior probabilities under H_0 and H_1

$$P(\theta \in A|x) \text{ and } P(\theta \in \Theta \setminus A|x)$$

- Reject H_0 if $P(H_1|x) > 0.5$ or some larger cutoff
- Problematic if H_0 is a point null hypothesis and θ is a continuous parameter

Bayes Factors

- Suppose we have two candidate hypotheses/models, H_0 and H_1 , with respective parameters space Θ_0 and Θ_1
- The Bayes factor (BF) in favor of H_0 over H_1

$$\text{BF} = \frac{\pi(\Theta_0|\mathbf{x})/\pi(\Theta_1|\mathbf{x})}{\pi(\Theta_0)/\pi(\Theta_1)},$$

where $\pi(\Theta_0|\mathbf{x})/\pi(\Theta_1|\mathbf{x})$ is the posterior odds and $\pi(\Theta_0)/\pi(\Theta_1)$ is the prior odds

- Only defined when the prior is proper, i.e. $\int_{\Theta_0} \pi(\theta)d\theta < \infty$ and $\int_{\Theta_1} \pi(\theta)d\theta < \infty$

Bayes Factors

- The BF can be also written as the ratio of the observed marginal densities

$$\begin{aligned}\text{BF} &= \frac{\pi(\Theta_0|\mathbf{x})/\pi(\Theta_1|\mathbf{x})}{\pi(\Theta_0)/\pi(\Theta_1)} \\ &= \frac{\left(\frac{\pi(\mathbf{x}|\Theta_0)\pi(\Theta_0)}{\pi(\mathbf{x})}\right) / \left(\frac{\pi(\mathbf{x}|\Theta_1)\pi(\Theta_1)}{\pi(\mathbf{x})}\right)}{\pi(\Theta_0)/\pi(\Theta_1)} \\ &= \frac{\pi(\mathbf{x}|\Theta_0)}{\pi(\mathbf{x}|\Theta_1)} \\ &= \frac{\int_{\Theta_0} f(\mathbf{x}|\theta)\pi(\theta)d\theta}{\int_{\Theta_1} f(\mathbf{x}|\theta)\pi(\theta)d\theta}\end{aligned}$$

- Interpretation: How likely are the data based on each model, integrating over the uncertainty in the parameters

Bayes Factors

- The Bayes factor is a ratio of the marginal likelihoods under H_0 and H_1
- Provides magnitude of evidence for H_0 over H_1
- Reject H_0 if BF is “small”
- How small is “small” and how large is “large”

Bayes Factors

- Jeffreys' rule

$\log_{10} \text{BF}$	BF	Strength of evidence
$[0, 1/2]$	$[1, 3.2]$	Barely worth mentioning
$(1/2, 1]$	$(3.2, 10]$	Substantial
$(1, 2]$	$(10, 100]$	Strong
$(2, \infty)$	$(100, \infty]$	Decisive

- Kass and Raftery' rule

$2 \log_e \text{BF}$	BF	Strength of evidence
$[0, 2]$	$[1, 3]$	Barely worth mentioning
$(2, 6]$	$(3, 20]$	Substantial
$(6, 10]$	$(20, 150]$	Strong
$(10, \infty)$	$(150, \infty]$	Decisive

In-class Practice

- Calculate the Bayes factor $B(H_0, H_1)$ in Example 1 and 2

Example 1

- A treatment is tested on 21 patients and it is found to have a positive effect in 18 patients. What is the probability that the treatment is effective?

$$x_1, \dots, x_n | \omega \sim \text{Bern}(\omega)$$

- Denote X_i as the binary outcome of patient i after the treatment with $X_i = 1$ for effective
- Sample size $n = 21$
- Sample sum $x_{\cdot} = \sum_{i=1}^n x_i = 18$
- Effective rate ω

- Example 1: Is the effective rate greater than 0.5?
 - Identify the null and alternative hypotheses

$$H_0 : \omega \leq 0.5 \text{ vs. } H_1 : \omega > 0.5$$

- Specify a prior $\omega \sim \text{Beta}(a = 1, b = 1)$
- Derive the posterior $\omega|\mathbf{x} \sim \text{Beta}(a + x_., b + n - x_.)$
- Calculate the Bayes Factor

$$\text{BF} = \frac{\pi(\omega \leq 0.5|\mathbf{x})/\pi(\omega > 0.5|\mathbf{x})}{\pi(\omega \leq 0.5)/\pi(\omega > 0.5)}$$

- Exact solution

```
# Read data information
n <- 21
x_sum <- 18

# Set the prior setting
a <- 1
b <- 1

# Calculate posterior probabilities
Theta_0_post <- pbeta(q = 0.5, shape1 = a + x_sum, shape2 = b + n - x_sum)
Theta_1_post <- 1 - Theta_0_post
```

- Exact solution

```
# Calculate prior probabilities
Theta_0_prior <- pbeta(q = 0.5, shape1 = a, shape2 = b)
Theta_1_prior <- 1 - Theta_0_prior

# Obtain the Bayes Factor on log scale -15.51
print(round(2*log(Theta_0_post/Theta_1_post)/
(Theta_0_prior/Theta_1_prior), 2))

# Obtain the Bayes Factor on log scale 15.51
print(round(2*log(Theta_1_post/Theta_0_post)/
(Theta_1_prior/Theta_0_prior), 2))
```

- Solutions (simulation-based)

```
# Set the algorithm setting
U <- 10000
set.seed(1)

# Simulate samples from the prior
omega_prior <- rbeta(U, shape1 = a, shape2 = b)

# Simulate samples from the posterior
omega_post <- rbeta(U, shape1 = a + x_sum, shape2 = b + n - x_sum)
```

- Solutions (simulation-based)

```
# Calculate posterior probabilities
Theta_0_post <- sum(omega_post <= 0.5)/U
Theta_1_post <- sum(omega_post > 0.5)/U

# Calculate prior probabilities
Theta_0_prior <- sum(omega_prior <= 0.5)/U
Theta_1_prior <- sum(omega_prior > 0.5)/U

# Obtain the Bayes Factor on log scale 14.44
print(round(2*log(Theta_1_post/Theta_0_post)/
(Theta_1_prior/Theta_0_prior), 2))
```

Example 2

- A study in Germany found that of a 980 placenta previa births, 437 were female. What is the proportion of female births in the population of women with placenta previa?

$$x_1, \dots, x_n | \omega \sim \text{Bern}(\omega)$$

- Denote X_i as the gender of birth i with $X_i = 1$ for female
- Sample size $n = 980$
- Sample sum $x_{\cdot} = \sum_{i=1}^n x_i = 437$
- The female birth proportion ω

In-class Practice

- Example 2: Is the male:female ratio at birth is significantly higher (i.e. ω is significantly lower) in women with placenta previa than in those without placenta previa?
 - Identify the null and alternative hypotheses

$$H_0 : \omega_x \geq \omega_y \text{ vs. } H_1 : \omega_x < \omega_y$$

- Specify priors $\omega_x, \omega_y \sim \text{Beta}(a = 1, b = 1)$
- Derive the posteriors $\omega_x | \mathbf{x} \sim \text{Beta}(a + x_., b + n_x - x_.)$ and $\omega_y | \mathbf{y} \sim \text{Beta}(a + y_., b + n_y - y_.)$
- Calculate the Bayes Factor

$$\text{BF} = \frac{\pi(\omega_x \geq \omega_y | \mathbf{x}, \mathbf{y}) / \pi(\omega_x < \omega_y | \mathbf{x}, \mathbf{y})}{\pi(\omega_x \geq \omega_y) / \pi(\omega_x < \omega_y)}$$

- Solutions (simulation-based)

```
# Read data information
n_x <- 980
x_sum <- 437
n_y <- 820
y_sum <- 398

# Set the prior setting
a <- 1
b <- 1

# Set the algorithm setting
U <- 10000
set.seed(1)
```


- Solutions (simulation-based)

```
# Simulate samples from priors
omega_x_prior <- rbeta(U, shape1 = a, shape2 = b)
omega_y_prior <- rbeta(U, shape1 = a, shape2 = b)

# Simulate samples from posteriors
omega_x_post <- rbeta(U, shape1 = a + x_sum, shape2 = b + n_x - x_sum)
omega_y_post <- rbeta(U, shape1 = a + y_sum, shape2 = b + n_y - y_sum)
```

- Solutions (simulation-based)

```
# Calculate posterior probabilities
Theta_0_post <- sum(omega_x_post >= omega_y_post)/U
Theta_1_post <- sum(omega_x_post < omega_y_post)/U

# Calculate prior probabilities
Theta_0_prior <- sum(omega_x_prior >= omega_y_prior)/U
Theta_1_prior <- sum(omega_x_prior < omega_y_prior)/U
```

- Solutions (simulation-based)

```
# Obtain the Bayes Factor on log scale -5.96
print(round(2*log(Theta_0_post/Theta_1_post)/
(Theta_0_prior/Theta_1_prior), 2))

# Obtain the Bayes Factor on log scale 5.96
print(round(2*log(Theta_1_post/Theta_0_post)/
(Theta_1_prior/Theta_0_prior), 2))
```

Example 3: Basketball Free Throws

A basketball player makes a free throw with probability ω . Suppose the player makes 18 of 25 free throws and assume

$$\omega \sim \text{Beta}(2, 2).$$

- 1 Derive the posterior distribution of ω .
- 2 Calculate the posterior mean

$$E(\omega \mid x).$$

- 3 Calculate the posterior probability

$$P(\omega > 0.70 \mid x).$$

- 4 Find the posterior predictive probability that the player makes the next free throw.

Use the posterior distribution to answer all four questions.

Example 3: Basketball Free Throws

We have

$$P(X = 8) = \binom{10}{8} (0.75)^8 (0.25)^2.$$

For at least 8 makes,

$$P(X \geq 8) = P(X = 8) + P(X = 9) + P(X = 10).$$

The expected number of makes is

$$E[X] = 10(0.75) = 7.5.$$

The expected value does not need to be an integer.

Example 3: Basketball in R

Calculate

```
dbinom(8, size = 10, prob = 0.75)
```

Calculate the probability of at least 8 makes:

```
1 - pbinom(7, size = 10, prob = 0.75)
```

Calculate the mean:

```
10 * 0.75
```

Example 4: Baseball Hitting

Suppose a baseball player has an unknown probability ω of getting a hit in an at-bat.

In 20 at-bats, the player gets

$$x_{\cdot} = 7$$

hits.

Suppose

$$\omega \sim \text{Beta}(2, 3).$$

Find:

- 1 the posterior distribution;
- 2 the posterior mean;
- 3 how the posterior mean compares with the sample proportion.

Example 4: Baseball Hitting

The posterior is

$$\omega \mid \mathbf{x} \sim \text{Beta}(2 + 7, 3 + 20 - 7).$$

Therefore,

$$\omega \mid \mathbf{x} \sim \text{Beta}(9, 16).$$

The posterior mean is

$$E[\omega \mid \mathbf{x}] = \frac{9}{25} = 0.36.$$

The sample proportion is

$$\bar{x} = \frac{7}{20} = 0.35.$$

The prior mean is

Example 5: Tennis Serve Percentage

A tennis player has made

38

first serves out of

50

attempts.

Let

ω = probability of a successful first serve.

Suppose

$$\omega \sim \text{Beta}(10, 10).$$

Find:

- 1 the posterior distribution;

Example 5: Tennis Serve Percentage

The posterior distribution is

$$\omega \mid \mathbf{x} \sim \text{Beta}(48, 22).$$

The posterior mean is

$$E[\omega \mid \mathbf{x}] = \frac{48}{70} \approx 0.686.$$

Compare this with the sample proportion:

$$\frac{38}{50} = 0.76.$$

The posterior mean is pulled toward the prior mean 0.50.

Example 6: Same Data, Different Priors

Suppose

$$x_{\cdot} = 7, \quad n = 20.$$

Consider three priors:

$$\text{Beta}(1, 1), \quad \text{Beta}(2, 2), \quad \text{Beta}(20, 20).$$

The posterior mean is

$$E[\omega \mid x] = \frac{a + x_{\cdot}}{a + b + n}.$$

Calculate the three posterior means.

Example 6: Prior Sensitivity

For Beta(1, 1),

$$E[\omega \mid x] = \frac{8}{22} \approx 0.364.$$

For Beta(2, 2),

$$E[\omega \mid x] = \frac{9}{24} = 0.375.$$

For Beta(20, 20),

$$E[\omega \mid x] = \frac{27}{60} = 0.450.$$

A stronger prior pulls the posterior more strongly toward its prior mean.

Example 7: Prior as Pseudo-Information

Recall

$$E[\omega \mid \mathbf{x}] = \frac{a + x.}{a + b + n}.$$

Rewrite:

$$E[\omega \mid \mathbf{x}] = \frac{a + b}{a + b + n} \frac{a}{a + b} + \frac{n}{a + b + n} \frac{x.}{n}.$$

Thus $a + b$ acts like a measure of the prior's effective sample size.

For example,

Beta(20, 20)

has prior mean 0.50 and effective sample size 40.

Example 8: A Small Sample

A basketball player makes

1

of

2

three-point attempts.

The sample proportion is

$$\hat{\omega} = 0.5.$$

Suppose

$$\omega \sim \text{Beta}(10, 10).$$

Then

$$10 + 1$$

Example 9: Same Prior, More Data

Now suppose the player makes

70

of 100 attempts.

The same prior is used:

$$\omega \sim \text{Beta}(10, 10).$$

The posterior mean is

$$E[\omega \mid x] = \frac{10 + 70}{10 + 10 + 100} = \frac{2}{3} \approx 0.667.$$

The sample proportion is

$$\frac{70}{100} = 0.70.$$

The larger sample gives the data more influence.

Example 10: How Does Sample Size Affect Uncertainty?

Suppose

$$\omega \sim \text{Beta}(1, 1).$$

Compare

$$n = 10, \quad x_{\cdot} = 5$$

with

$$n = 100, \quad x_{\cdot} = 50.$$

Both have

$$\bar{x} = 0.5.$$

But the posterior distributions are

$$\text{Beta}(6, 6)$$

Example 10: Posterior Precision

For a $\text{Beta}(a, b)$ distribution,

$$\text{Var}(\omega) = \frac{ab}{(a+b)^2(a+b+1)}.$$

Thus

$$\text{Var}\{\text{Beta}(6, 6)\} = \frac{36}{12^2(13)},$$

whereas

$$\text{Var}\{\text{Beta}(51, 51)\} = \frac{51^2}{102^2(103)}.$$

More data reduce posterior uncertainty.

Example 11: Posterior Probability

Suppose

$$\omega \mid x \sim \text{Beta}(19, 4).$$

Ask

$$P(\omega > 0.5 \mid x) = ?$$

This is a probability statement about the unknown parameter.

It directly asks how probable it is, under the Bayesian model, that the success rate exceeds 0.5 after seeing the data.

Example 11: Posterior Probability in R

Use the Beta CDF:

```
1 - pbeta(  
  0.5,  
  shape1 = 19,  
  shape2 = 4  
)
```

Or use Monte Carlo:

```
set.seed(1)  
omega <- rbeta(10000, 19, 4)  
mean(omega > 0.5)
```

The two approaches should give similar answers.

Example 12: Predicting the Next Free Throw

Suppose

$$\omega \mid x \sim \text{Beta}(19, 4).$$

Let X_{new} indicate whether the next free throw is made.

We want

$$P(X_{\text{new}} = 1 \mid x).$$

By the law of total probability,

$$P(X_{\text{new}} = 1 \mid x) = \int P(X_{\text{new}} = 1 \mid \omega) \pi(\omega \mid x) d\omega.$$

Example 12: Predictive Probability

Because

$$P(X_{\text{new}} = 1 \mid \omega) = \omega,$$

we have

$$P(X_{\text{new}} = 1 \mid x) = E[\omega \mid x].$$

Therefore,

$$P(X_{\text{new}} = 1 \mid x) = \frac{19}{23} \approx 0.826.$$

Inference about a parameter leads directly to prediction.

Example 13: Predicting Multiple Free Throws

Suppose

$$\omega \mid x \sim \text{Beta}(19, 4).$$

Let

Y_{new} = number of makes in 5 future free throws.

Conditional on ω ,

$$Y_{\text{new}} \mid \omega \sim \text{Bin}(5, \omega).$$

Question:

$$P(Y_{\text{new}} = 4 \mid x) = ?$$

Example 13: Beta-Binomial Prediction

We integrate over posterior uncertainty:

$$P(Y_{\text{new}} = 4 \mid x) = \int P(Y_{\text{new}} = 4 \mid \omega) \pi(\omega \mid x) d\omega.$$

Thus,

$$P(Y_{\text{new}} = 4 \mid x) = \binom{5}{4} \frac{B(19 + 4, 4 + 5 - 4)}{B(19, 4)}.$$

Therefore,

$$Y_{\text{new}} \mid x \sim \text{BetaBinomial}(5, 19, 4).$$

In-class Practice: Monte Carlo

Suppose

$$\omega \mid x \sim \text{Beta}(12, 8).$$

Generate 10,000 posterior draws.

Use them to estimate

$$P(\omega > 0.6 \mid x).$$

Then estimate

$$E[\omega^2 \mid x].$$

Finally, compare two independent posterior distributions and estimate

$$P(\omega_1 > \omega_2 \mid x_1, x_2).$$

In-class Practice: Monte Carlo, Solution

Use posterior simulation.

```
set.seed(1)
U <- 10000

omega <- rbeta(U, shape1 = 12, shape2 = 8)

# Posterior probability
mean(omega > 0.6)

# Posterior expectation of omega^2
mean(omega^2)

# Compare with a second posterior distribution
omega_2 <- rbeta(U, shape1 = 10, shape2 = 10)
mean(omega > omega_2)
```

Analytically,

$$E(\omega^2 \mid x) = \frac{12(13)}{20(21)} \approx 0.371.$$

In-class Practice: Functions of Parameters

Suppose

$$\omega \mid x \sim \text{Beta}(18, 12).$$

Let

$$g(\omega) = \frac{\omega}{1 - \omega}.$$

Use simulation to estimate

$$E[g(\omega) \mid x].$$

Also estimate the median of

$$g(\omega).$$

Transform every posterior draw, rather than deriving a new density.

In-class Practice: Functions of Parameters, Solution

Transform each posterior draw.

```
set.seed(1)
U <- 10000

omega <- rbeta(U, shape1 = 18, shape2 = 12)
g <- omega / (1 - omega)

mean(g)
median(g)
quantile(g, c(0.025, 0.5, 0.975))
```

Because $\omega \sim \text{Beta}(18, 12)$,

$$E\left(\frac{\omega}{1-\omega} \mid x\right) = \frac{18}{12-1} \approx 1.636.$$

Example 14: Comparing Two Basketball Players

Player A:

$$X_A = 8, \quad n_A = 10.$$

Player B:

$$X_B = 6, \quad n_B = 10.$$

Assume independent priors

$$\omega_A, \omega_B \sim \text{Beta}(1, 1).$$

Then

$$\omega_A \mid x_A \sim \text{Beta}(9, 3)$$

and

$$\omega_B \mid x_B \sim \text{Beta}(7, 5).$$

Example 14: Monte Carlo Comparison

Generate independent posterior draws

$$\omega_A^{(s)} \sim \text{Beta}(9, 3), \quad s = 1, \dots, S,$$

and

$$\omega_B^{(s)} \sim \text{Beta}(7, 5), \quad s = 1, \dots, S.$$

Define

$$q = P(\omega_A > \omega_B \mid x_A, x_B).$$

Then the Monte Carlo estimator is

$$\hat{q} = \frac{1}{S} \sum_{s=1}^S \mathbf{1} \left\{ \omega_A^{(s)} > \omega_B^{(s)} \right\}.$$

This can be easier than deriving the distribution of $\omega_A - \omega_B$ analytically.

Example 15: Interpreting a 95% Credible Interval

Suppose

$$\omega \mid x$$

has a 95% credible interval

$$[0.62, 0.81].$$

The Bayesian interpretation is

$$P(0.62 \leq \omega \leq 0.81 \mid x) = 0.95.$$

The probability statement concerns the unknown parameter, conditional on the observed data and the specified model.

In-class Practice: Confidence vs. Credible Interval

Suppose a frequentist analysis reports

$$95\% \text{ CI} = [0.62, 0.81]$$

and a Bayesian analysis reports

$$95\% \text{ CrI} = [0.62, 0.81].$$

The numerical intervals happen to be identical.

Are the interpretations identical?

No.

Discuss what is random in each framework.

In-class Practice: Confidence vs. Credible Interval, Solution

The numerical intervals may be the same, but the interpretations differ.

- Bayesian credible interval:

$$P(0.62 \leq \omega \leq 0.81 \mid x) = 0.95.$$

The parameter is treated as unknown and random conditional on the data.

- Frequentist confidence interval: In repeated sampling, 95% of intervals constructed by the method would contain the fixed parameter.

Same endpoints do not imply the same probability statement.

Example 16: MLE versus Posterior Mean

Suppose

$$x_{\cdot} = 3, \quad n = 10.$$

The MLE is

$$\hat{\omega}_{\text{MLE}} = 0.30.$$

Suppose

$$\omega \sim \text{Beta}(1, 1).$$

Then

$$E[\omega \mid x] = \frac{1 + 3}{1 + 1 + 10} = \frac{1}{3} \approx 0.333.$$

The difference comes from incorporating prior information.

Example 17: MAP versus MLE

Suppose

$$x_{\cdot} = 3, \quad n = 10,$$

and

$$\omega \sim \text{Beta}(5, 5).$$

The posterior is

$$\omega \mid x \sim \text{Beta}(8, 12).$$

The posterior mode is

$$\frac{8-1}{8+12-2} = \frac{7}{18} \approx 0.389.$$

Compare this with

$$\hat{\omega}_{\text{MLE}} = 0.30.$$

Example 18: Near the Boundary

Suppose

$$x_{\cdot} = 1, \quad n = 10,$$

and

$$\omega \sim \text{Beta}(1, 1).$$

Then

$$\omega \mid x \sim \text{Beta}(2, 10).$$

The posterior is strongly concentrated near zero and is skewed.
A normal approximation can assign probability to

$$\omega < 0,$$

which is impossible.

Exact calculations or transformations may be preferable.

Example 19: Logit Transformation

For

$$0 < \omega < 1,$$

define

$$\eta = \log \left(\frac{\omega}{1 - \omega} \right).$$

Then

$$\eta \in \mathbb{R}.$$

The inverse transformation is

$$\omega = \frac{e^{\eta}}{1 + e^{\eta}}.$$

A skewed posterior on the ω scale may be closer to symmetric on the logit

Integrated Example: Basketball Player

A basketball player makes 18 of 25 free throws.

Suppose

$$\omega \sim \text{Beta}(2, 2).$$

Find:

- 1 the posterior distribution;
- 2 the posterior mean;
- 3 the posterior variance;
- 4 a 95% credible interval;
- 5 $P(\omega > 0.70 \mid x)$;
- 6 the probability that the player makes the next free throw.

Integrated Example: Solution

The posterior is

$$\omega \mid x \sim \text{Beta}(20, 9).$$

The posterior mean is

$$E[\omega \mid x] = \frac{20}{29} \approx 0.690.$$

The posterior variance is

$$\text{Var}(\omega \mid x) = \frac{20(9)}{29^2(30)}.$$

The posterior predictive probability of making the next free throw is

$$P(X_{\text{new}} = 1 \mid x) = \frac{20}{29}.$$

In-class Practice: Prior Sensitivity

Suppose

$$X \mid \omega \sim \text{Bin}(50, \omega)$$

and $x = 28$.

Compare the priors

$$\text{Beta}(1, 1), \quad \text{Beta}(5, 5), \quad \text{Beta}(20, 5).$$

For each prior:

- 1 derive the posterior;
- 2 compute the posterior mean;
- 3 determine whether the prior pulls the posterior above or below the sample proportion.

In-class Practice: Prior Sensitivity, Solution

The sample proportion is

$$\bar{x} = \frac{28}{50} = 0.56.$$

Prior	Posterior	Posterior mean	Direction
Beta(1, 1)	Beta(29, 23)	$29/52 = 0.558$	slightly below 0.56
Beta(5, 5)	Beta(33, 27)	$33/60 = 0.550$	below 0.56
Beta(20, 5)	Beta(48, 27)	$48/75 = 0.640$	above 0.56

The posterior mean is pulled toward the prior mean, with stronger pull when $a + b$ is larger.

In-class Practice: Rare Event

Suppose a baseball player has

$$X = 1$$

hit in

$$n = 20$$

at-bats. Assume

$$\omega \sim \text{Beta}(1, 1).$$

Compare

$$P(\omega < 0.10 \mid x) \text{ and } P(\omega > 0.10 \mid x).$$

How would your answer change under the prior

$$\text{Beta}(1, 9)?$$

In-class Practice: Rare Event, Solution

With the uniform prior,

$$\omega \mid x \sim \text{Beta}(2, 20).$$

Thus,

$$P(\omega < 0.10 \mid x) \approx 0.635, \quad P(\omega > 0.10 \mid x) \approx 0.365.$$

With the Beta(1, 9) prior,

$$\omega \mid x \sim \text{Beta}(2, 28),$$

so

$$P(\omega < 0.10 \mid x) \approx 0.801.$$

```
pbeta(0.10, shape1 = 2, shape2 = 20)
pbeta(0.10, shape1 = 2, shape2 = 28)
```

In-class Practice: Two Groups

Two basketball players have independent success probabilities ω_A and ω_B .

Player A makes 16 of 20 shots.

Player B makes 12 of 20 shots.

Assume

$$\omega_A, \omega_B \sim \text{Beta}(1, 1).$$

- 1 Find both posterior distributions.
- 2 Compare posterior means.
- 3 Estimate $P(\omega_A > \omega_B \mid x_A, x_B)$ by simulation.

In-class Practice: Two Groups, Solution

The posteriors are

$$\omega_A \mid x_A \sim \text{Beta}(17, 5), \quad \omega_B \mid x_B \sim \text{Beta}(13, 9).$$

The posterior means are

$$E(\omega_A \mid x_A) = \frac{17}{22} = 0.773, \quad E(\omega_B \mid x_B) = \frac{13}{22} = 0.591.$$

Estimate the comparison probability by simulation.

```
set.seed(1)
U <- 10000
omega_A <- rbeta(U, shape1 = 17, shape2 = 5)
omega_B <- rbeta(U, shape1 = 13, shape2 = 9)
mean(omega_A > omega_B)
```

The Monte Carlo estimate is about 0.91.

In-class Practice: Posterior Prediction

Suppose

$$\omega \mid x \sim \text{Beta}(18, 7).$$

A player will take 10 future shots.

Let

$$Y^* \mid \omega \sim \text{Bin}(10, \omega).$$

Find or estimate:

- ① $P(Y^* = 8 \mid x);$
- ② $P(Y^* \geq 8 \mid x);$
- ③ $E[Y^* \mid x].$

Compare the predictive mean with

$$10E[\omega \mid x].$$

In-class Practice: Posterior Prediction, Solution

The posterior predictive distribution is beta-binomial:

$$P(Y^* = y \mid x) = \binom{10}{y} \frac{B(18 + y, 7 + 10 - y)}{B(18, 7)}.$$

Therefore,

$$P(Y^* = 8 \mid x) \approx 0.231, \quad P(Y^* \geq 8 \mid x) \approx 0.462.$$

The predictive mean is

$$E(Y^* \mid x) = 10E(\omega \mid x) = 10\frac{18}{25} = 7.2.$$

```
alpha <- 18
beta <- 7
m <- 10

choose(m, 8) * beta(alpha + 8, beta + m - 8) / beta(alpha, beta)
```

After-class Reading

- *A First Course in Bayesian Statistical Methods* by Peter Hoff
 - Chapter 3 One-parameter models
 - Section 3.1 The binomial model